MA 102 Mathematics II

Ordinary Differential Equations

Lecture 19

(April 22, 2024)

A steady state does not have much significance unless it is stable. Consider the case of a pendulum: there are two possible steady states - (i) when the bob is at rest at the highest point (corresponding to maximum displacement), (ii) when the bob is at rest at the lowest point (corresponding to zero displacement).

The first state is clearly unstable, and the second one is stable. Recall that a steady state of a simple physical system corresponds to an equilibrium point (or critical point) in the phase plane.

These considerations suggest in a general way that a small disturbance at an unstable equilibrium point leads to a larger and larger departure from this point while the opposite is true at a stable equilibrium point.

In order to discuss it in a more mathematical way, consider an isolated critical point of the system

$$\left.\begin{aligned} \frac{dx}{dt} &= F(x,y), \\ \frac{dy}{dt} &= G(x,y), \end{aligned}\right\} \tag{1}$$

and assume that the origin O$=(0,0)$ is this critical point of the phase plane.

This critical point is said to be stable if, for each positive number R, there exists a positive number $r \leq R$ such that every path which is inside the circle $x^2 + y^2 = r^2$ for some $t = t_0$ remains inside the circle $x^2 + y^2 = R^2$ for all $t > t_0$. (Figure 1)

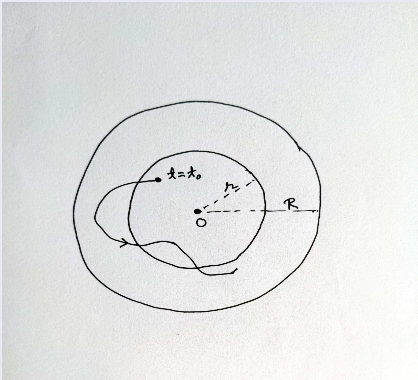

Figure 1

We can also say that a critical point is stable if all paths that get sufficiently close to the point stay close to the point. Further, the critical point is said to be asymptotically stable if it is stable and there exists a circle $x^2 + y^2 = r_0^2$ such that every path which is inside this circle for some $t = t_0$ approaches the origin as $t \to \infty$.

If the critical point is not stable, it is called an unstable one.

From the figures in the last lecture, we may notice that

- The node in Figure 2, the saddle point in Figure 3 and the spiral on the left in Figure 6 are unstable.
- The center in Figure 4 is stable but not asymptotically stable.
- The node in Figure 1, the spiral in Figure 5 and the spiral on the right in Figure 6 are asymptotically stable.

Having studied different types of critical points, we now have some information how to conclude what critical point a system possesses. Now we examine how to ascertain the type of critical point by thoroughly analyzing the nature of the roots including their signs.

Recall

Consider the following linear autonomous system:

$$\left.\begin{aligned} \frac{dx}{dt} &= a_1 x + b_1 y, \\ \frac{dy}{dt} &= a_2 x + b_2 y, \end{aligned}\right\} \tag{2}$$

which has $(0,0)$ as an obvious critical point.

We assume throughout that

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0 \tag{3}$$

so that $(0,0)$ is the only critical point.

(2) has a non-trivial solution of the form

$$\left.\begin{aligned} x &= Ae^{mt}, \\ y &= Be^{mt}. \end{aligned}\right\} \tag{4}$$

The solutions (4) are obtained whenever m is a root of the quadratic equation

$$m^2 - (a_1 + b_2)m + (a_1 b_2 - a_2 b_1) = 0, \tag{5}$$

called the auxiliary equation of the system.
We may note that condition (3) implies that zero cannot be a root of (5).

Let m_1 and m_2 be the roots of (5). The nature of the critical point $(0, 0)$ of the system (2) is determined by the nature of the numbers m_1 and m_2 as follows (this time it is not just three cases as usual but five):

Major Cases:

- **Case A:** The roots m_1, m_2 are real, distinct and of the same sign implies $(0,0)$ is a node.
- **Case B:** The roots m_1, m_2 are real, distinct and of opposite sign implies $(0,0)$ is a saddle point.
- **Case C:** The roots m_1, m_2 are complex conjugates but are not purely imaginary implies $(0,0)$ is a spiral.

Minor Cases:

- **Case D:** The roots m_1, m_2 are real and equal implies $(0,0)$ is a node.
- **Case E:** The roots m_1, m_2 are purely imaginary implies $(0,0)$ is a center.

Theorem (Case A)

If the roots m_1 and m_2 of (5) are real, distinct, and of the same sign, then the critical point $(0,0)$ of the system (2) is a node.

Proof: Assume that both m_1 and m_2 are negative and choose $m_1 < m_2 < 0$.

The general solution of (2) has the form

$$\left.\begin{aligned} x &= c_1 A_1 e^{m_1 t} + c_2 A_2 e^{m_2 t}, \\ y &= c_1 B_1 e^{m_1 t} + c_2 B_2 e^{m_2 t}, \end{aligned}\right\} \tag{6}$$

where the A's and B's are definite constants such that $B_1/A_1 \neq B_2/A_2$, and the c's are arbitrary constants.

When $c_2 = 0$, we have the solutions as

$$\left.\begin{aligned} x &= c_1 A_1 e^{m_1 t}, \\ y &= c_1 B_1 e^{m_1 t}. \end{aligned}\right\} \tag{7}$$

When $c_1 = 0$, we have the solutions as

$$\left.\begin{aligned} x &= c_2 A_2 e^{m_2 t}, \\ y &= c_2 B_2 e^{m_2 t}. \end{aligned}\right\} \tag{8}$$

For any $c_1 > 0$, the solutions (7) represent a path consisting of half of the line $A_1 y = B_1 x$ with slope B_1/A_1 while for any $c_1 < 0$, the solution (7) represents a path consisting of the other half of this line (on the other side of the origin).

Since $m_1 < 0$, both of these half-line paths approach $(0,0)$ as $t \to \infty$, and since $y/x = B_1/A_1$, both enter $(0,0)$ with slope B_1/A_1.

Similarly, in exactly the similar way, the solutions (8) represent two half-line paths lying on the line $A_2 y = B_2 x$ with slope B_2/A_2. These two paths also approach $(0,0)$ as $t \to \infty$, and enter $(0,0)$ it with slope B_2/A_2.

Now, for the case $c_1 \neq 0$ and $c_2 \neq 0$, the general solution (6) represents curved paths. Since $m_1 < 0$ and $m_2 < 0$, these paths also approach $(0,0)$ as $t \to \infty$.

Further, we know that $m_1 - m_2 < 0$, and have

$$\frac{y}{x} = \frac{c_1 B_1 e^{m_1 t} + c_2 B_2 e^{m_2 t}}{c_1 A_1 e^{m_1 t} + c_2 A_2 e^{m_2 t}} = \frac{(c_1 B_1/c_2) e^{(m_1 - m_2)t} + B_2}{(c_1 A_1/c_2) e^{(m_1 - m_2)t} + A_2}.$$

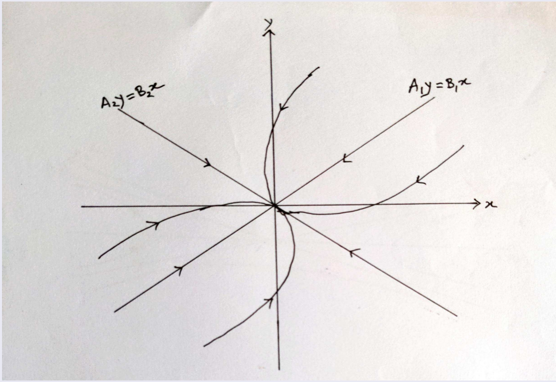

Figure 2

It is evident that $y/x \to B_2/A_2$ as $t \to \infty$ and therefore, all of these paths enter $(0,0)$ with slope B_2/A_2.

It is clear that the critical point is a node and it is asymptotically stable.

If m_1 and m_2 are both positive and choose them such that $m_1 > m_2 > 0$, then the situation is exactly the same except that all the paths now approach and enter $(0,0)$ as $t \to -\infty$. In that case, the paths will remain the same as in Figure 2 but the arrows reverse the direction. $(0,0)$ still remains as a node but it is unstable this time.

Theorem (Case B)

If the roots m_1 and m_2 of (5) are real, distinct, and of opposite signs, then the critical point $(0,0)$ of the system (2) is a saddle point.

Proof

Here, as per the given condition, we can take $m_1 < 0 < m_2$. The general solution of (2) can still be written in the form (6). Corresponding to the roots, we have the particular solutions in the forms (7) and (8).

The two half-line paths represented by (7) still approach and enter $(0,0)$ as $t \to \infty$ but the two half-line paths represented by (8) approach and enter $(0,0)$ when $t \to -\infty$ (against $t \to \infty$ for Case A).

If $c_1 \neq 0$ and $c_2 \neq 0$, the general solution (6) still represents curved paths but since $m_1 < 0 < m_2$, none of these paths approaches $(0,0)$ as $t \to \infty$ or $t \to -\infty$

Instead, as $t \to \infty$, each of these curves is asymptotic to one of the half-line paths represented by (8), and as $t \to -\infty$, each of these curves is asymptotic to one of the half-line paths represented by (7).
Refer to Figure 3 on the next slide.

In this case, the critical point is a saddle point and it is obviously unstable.

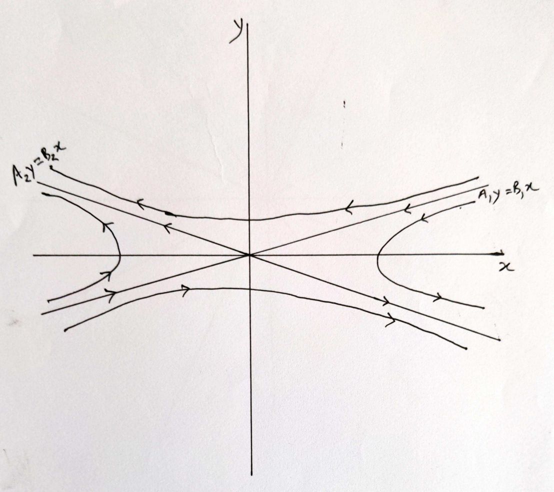

Figure 3

Theorem (Case C)

If the roots m_1 and m_2 of (5) are complex conjugates but not purely imaginary, then the critical point $(0, 0)$ of the system (2) is a spiral.

Proof

Here, we write the roots m_1 and m_2 in the form $a \pm \mathrm{i}b$, where $a \neq 0, b \neq 0$ are in $\mathbb{R}$. We also observe that the discriminant of (5) is negative:

$$
\begin{aligned}
D &= (a_1 + b_2)^2 - 4(a_1 b_2 - a_2 b_1) \qquad (9) \\
&= (a_1 - b_2)^2 + 4a_2 b_1 < 0. \qquad (10)
\end{aligned}
$$

The general solution of (2) is

$$
\left.
\begin{aligned}
x &= e^{at}[c_1(A_1 \cos bt - A_2 \sin bt) + c_2(A_1 \sin bt + A_2 \cos bt)], \\
y &= e^{at}[c_1(B_1 \cos bt - B_2 \sin bt) + c_2(B_1 \sin bt + B_2 \cos bt)],
\end{aligned}
\right\} \qquad (11)
$$

where the A's and B's are definite constants and the c's are arbitrary constants.

First let us assume that $a < 0$. Then it is clear from formulas (11) that $x \to 0$ and $y \to 0$ as $t \to \infty$. Now, it is required to prove that the paths do not enter the point $(0, 0)$ but wind around it in a spiral-like manner.

We introduce the polar coordinate θ and show that, along any path, $d\theta/dt$ is either positive for all t or negative for all t. We know $\theta = \arctan(y/x)$ and hence

$$\frac{d\theta}{dt} = \frac{x(dy/dt) - y(dx/dt)}{x^2 + y^2}.$$

Using the equations in the given system (2)

$$\frac{d\theta}{dt} = \frac{a_2x^2 + (b_2 - a_1)xy - b_1y^2}{x^2 + y^2}. \tag{12}$$

We are interested only in the solutions that represent paths and hence we assume that $x^2 + y^2 \neq 0$. From (10), we know that a_2 and b_1 have opposite signs and let us take $a_2 > 0$ and $b_1 < 0$.

When $y = 0$, we get from (12): $d\theta/dt = a_2 > 0$.

When $y \neq 0$, $d\theta/dt$ cannot be zero since if it is zero, we will get

$$a_2x^2 + (b_2 - a_1)xy - b_1y^2 = 0$$

i.e.,

$$a_2(x/y)^2 + (b_2 - a_1)(x/y) - b_1 = 0, \tag{13}$$

for some real number x/y which cannot hold because the discriminant of the quadratic equation (13) is negative due to (10).

This shows that $d\theta/dt$ is always positive when $a_2 > 0$.
In a similar manner, $d\theta/dt$ is always negative when $a_2 < 0$.

By (11), x and y change sign infinitely as $t \to \infty$ and therefore all paths must spiral in to the origin $-$ counter-clockwise if $a_2 > 0$ and clockwise if $a_2 < 0$.
The critical point in this case is therefore a spiral and it is asymptotically stable.

If $a > 0$, the situation remains the same except that the paths approach $(0,0)$ as $t \to -\infty$ and the critical point is unstable .
Refer to Figure 4 on the next slide (for $a_2 > 0$).

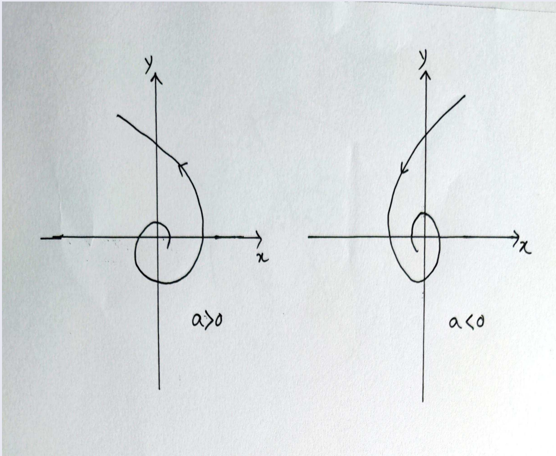

Figure 4